

Software Requirements Specification

For Railway Ticket Booking System

1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the **Railway Ticket Booking Platform System**. The purpose of this document is to provide a clear and detailed description of the system's behaviour, constraints, and interfaces to all stakeholders involved in its development and usage.

The scope of this SRS covers the complete online railway ticket booking process, including user registration and authentication, train search and seat availability checking, real-time train status tracking, ticket booking under different quotas (including Tatkal), online payment processing, and booking history management.

This document serves as a reference for developers, testers, designers, and project managers to ensure that the system is implemented in accordance with the defined requirements and conforms to Indian Railway reservation standards.

1.2 Document Conventions

This SRS document follows standard IEEE-style conventions for software requirement documentation.

- Requirements are written in clear, unambiguous, and testable language.
- Section headings and numbering follow a hierarchical structure for easy navigation.
- Priority levels for requirements are assumed to be uniform unless specified in later sections of the document.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- Developers: To understand system functionality, interfaces, and constraints before implementation.
- Project Managers: To plan development timelines, allocate resources, and track progress.
- Testers/QA Engineers: To design test cases based on functional and non-functional requirements.
- System Architects: To design scalable and reliable system architecture.
- End Users and Stakeholders: To understand the overall capabilities and limitations of the system.

Readers new to the project are advised to begin with Section 1 (Introduction) to gain a high-level understanding of the system. Technical readers may then proceed to the functional and non-functional requirements sections, while testers should focus on requirements and interface specifications.

1.4 Product Scope

The **Railway Ticket Booking Platform System** is a web- and mobile-based application designed to simplify and automate the railway ticket reservation process across India. The system aims to reduce dependency on physical ticket counters, minimize booking delays, and handle high traffic volumes during peak travel seasons.

Key objectives of the system include:

- Providing a seamless and user-friendly booking experience.
- Supporting real-time train schedules, seat availability, and live train status.
- Enabling secure online payments and instant booking confirmation.
- Supporting multiple booking quotas, including Tatkal.
- Ensuring high availability, scalability, and performance under heavy load.

The platform aligns with the broader goal of digital transformation in public transportation services and enhances accessibility for users across different regions and languages in India.

1.5 References

- 830-1998 - IEEE Recommended Practice for Software Requirements Specifications
- Payment Gateway API documentation (RBI-compliant)
- National Train Enquiry System (NTES) documentation
- Indian Railways Passenger Reservation System (PRS) standards

2. Overall Description

2.1 Product Perspective

The Railway Ticket Booking Platform System is a new, self-contained software system intended to provide an online alternative to conventional railway ticket booking methods, including physical reservation counters and partially automated legacy systems. The system operates as a centralized digital platform that enables users to search for trains, check seat availability, book tickets under various quotas, and track train status in real time.

The platform is designed to work in coordination with existing railway reservation infrastructure. It does not replace the core railway reservation databases but interacts with them to fetch and reflect updated information such as seat availability, booking status, and changes made at physical ticket booking offices. The system also integrates with external services such as payment gateways and train tracking services to deliver end-to-end booking functionality.

From a user perspective, the product acts as a single interface through which all booking-related activities can be performed without requiring physical presence at a railway station.

2.2 Product Functions

- User registration and login using phone number and email address
- Searching available trains based on travel date, route, and coach type
- Displaying seat availability for selected trains and travel dates
- Viewing real-time train schedules and live running status
- Booking railway tickets under applicable quotas, including Tatkal
- Processing secure online payments for ticket booking
- Generating booking confirmation within defined time limits
- Maintaining booking logs and history for user reference

2.3 User Characteristics

- Education level: The system's users must at the very least be fluent in English and know how to read.
- Technical proficiency: The user should feel at ease utilizing this general-purpose computer software.

2.4 Operating Env.

- The System will function well in Windows 7 or newer operating systems.
- The System will function well in devices of Android 11 or higher versions.
- The System will function in iOS 18 or higher versions.

2.5 Design and Implementation Constraints

Our system will ensure reliability, scalability too also with compliance.

- The application must support web, Android, and iOS platforms
- Backend communication must be implemented using REST APIs
- The system must comply with Indian Railways reservation standards and policies
- The platform must be deployed on cloud infrastructure to enable horizontal scalability
- The relational database used must support horizontal scaling to handle high traffic volumes
- System performance must meet defined response time and availability requirements

2.6 User Documentation

To ease the user experience, following documents can be provided along with system :

1. Online guide articles
2. Frequently Asked Questions (FAQ) for common issues ke liye

2.7 Assumptions and Dependencies

- Payment gateway will be taken from existing companies
- Train live schedule and booking data provided by railway systems of government
- Regional language support depends on translation books by other authors.

3. External Interface Requirements

3.1 User Interfaces

- The system will provide a user friendly interface with clear navigation
- The system will allow users to register and login using phone no. and email.
- The system will display available trains and seat availability based on selected date and coach type.
- The system will support multiple regional languages to ensure usability across India.
- The system will maintain booking logs that allow users to view past ticket bookings.

3.2 Hardware Interfaces

- Supporting devices are android/ iOS smartphones and Windows/Mac operating system.
- Cache 512 Kb
- RAM 400 MB
- Hard Disk drive 40 GB
- Device drivers (programs) required so that software programs can use these hardware

3.3 Software Interfaces

- The system will integrate with payment gateway APIs to enable online ticket payments.
- The system will integrate with GPS-based train tracking systems to fetch live train status.
- The system will interact with railway reservation systems to reflect updates made at ticket booking offices.
- The system backend will expose and consume REST-based APIs for all internal and external communication.
- The system will store booking and user data using a relational database that supports horizontal scalability

3.4 Communications Interfaces

- The system will use internet-based communication for all user interactions and data exchange.
- The system will communicate with external services using standard HTTP-based protocols.
- The system will ensure secure data transmission during login and payment transactions.
- The system will synchronize booking and train status data with external systems in near real time.

4.System Features

4.1 User Registration and Authentication

4.1.1 Description and Priority : (allows access to the platform.)

4.1.2 Functional Requirements :

The system will allow users to register using a phone number and email address.

The system will allow registered users to log in to the platform.

The system will restrict access to booking features for unauthenticated users.

4.2 Train Search and Seat Availability

4.2.1 Description and Priority : (allows access to the platform.)

4.2.2 Functional Requirements :

The system will allow users to search for available trains based on selected travel dates.

The system will display seat availability based on coach type.

4.3 Train Schedule and Live Status Tracking

4.3.1 Description and Priority:

Provides real-time train schedule and running status.

4.3.2 Functional Requirements:

The system will display train schedules for selected routes.

The system will show live train running status in real time.

4.4 Ticket Booking and Quota Management

4.4.1 Description and Priority:

Supports railway ticket booking under applicable reservation rules.

4.4.2 Functional Requirements:

The system will allow users to book tickets based on seat availability.

The system will handle quota-based ticket booking.

The system will allow Tatkal ticket booking as per applicable rules.

4.5 Online Payment Processing

4.5.1 Description and Priority:

Enables users to pay for ticket bookings online. Priority: High.

4.5.2 Functional Requirements:

The system will allow users to make online payments for ticket bookings.

The system will process payments using integrated payment gateway services.

4.6 Booking History and Logs

4.6.1 Description and Priority:

Allows users to view previous booking records. Priority: Medium.

4.6.2 Functional Requirements:

The system will maintain booking logs for each user.

The system will allow users to view past ticket bookings.

5. Other Non-Functional Requirements

5.1 Performance Requirements

- The system should load search results and seat availability within 3–5 seconds.
- It should handle high traffic during peak hours (e.g., festival and Tatkal bookings) without crashing.
- Payment and booking confirmation should process within reasonable time (under 10 seconds).
- Should support large numbers of concurrent users.

5.2 Safety Requirements

- System failures should not lead to duplicate bookings or incorrect payments.
- Backup mechanisms must be available for database recovery.
- Any failure during payment must either auto-refund or keep the transaction pending for user review.

5.3 Security Requirements

- User login data and payment details must be encrypted.
- System should protect against common cyber attacks (e.g., SQL injection, phishing, unauthorized access).
- Access to administrative features should require higher authentication.
- Personal user data must not be shared without consent.

5.4 Software Quality Attributes

- **Reliability:** System should run consistently without frequent downtime.
- **Usability:** Interface should be easy for general users to operate.
- **Scalability:** Should scale during high demand seasons.
- **Maintainability:** Updates and bug fixes should be easy to apply.
- **Availability:** System should be accessible 24×7 except during scheduled maintenance.

5.5 Business Rules

- Booking, cancellation, and refund policies must follow Indian Railway rules.
- Quota rules (e.g., Tatkal, General) must be enforced.
- Payment must be completed for booking to be confirmed.
- Tickets once booked should generate a booking record for user reference.

6. Other Requirements

- The system shall use a centralized relational database.
- The database shall store user, train schedule, seat availability, booking, and payment data.
- Data integrity and consistency shall be maintained during booking and payment transactions.
- The database shall support high read/write operations during peak booking hours.

Appendix A: Glossary

Term	Description
API	Application Programming Interface
NTES	National Train Enquiry System
PRS	Passenger Reservation System
RBI	Reserve Bank of India

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
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